

TE Curve Verification Pipeline Familywise ONNX Report - periodic_mlp_harmonic

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/` .

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory: `output/validation_checks/track2_familywise_onnx_report/periodic_mlp_harmonic/2026-07-08-09-46-27__track2_periodic_mlp_harmonic_familywise_onnx_report` ;
- summary YAML: `output/validation_checks/track2_familywise_onnx_report/periodic_mlp_harmonic/2026-07-08-09-46-27__track2_periodic_mlp_harmonic_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/periodic_mlp_harmonic/2026-07-08-09-46-27__track2_periodic_mlp_harmonic_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/periodic_mlp_harmonic/2026-07-08-09-46-27__track2_periodic_mlp_harmonic_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `periodic_mlp_harmonic` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_periodic_mlp_harmonic_fw_simplified_setpoints</code>	<code>2026-07-08-01-11-12_te_periodic_mlp_harmonic_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_periodic_mlp_harmonic_bw_simplified_setpoints</code>	<code>2026-07-08-01-19-22_te_periodic_mlp_harmonic_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_periodic_mlp_harmonic_global_simplified_setpoints</code>	<code>2026-07-08-01-02-50_te_periodic_mlp_harmonic_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/periodic_mlp_harmonic/forward/2026-07-08-01-11-12_te_periodic_mlp_harmonic_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/periodic_mlp_harmonic/forward/2026-07-08-01-11-12_te_periodic_mlp_harmonic_fw_simplified_setpoints/python/periodic_mlp-epoch=055-val_mae=0.00280280.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/periodic_mlp_harmonic/backward/2026-07-08-01-19-22_te_periodic_mlp_harmonic_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/periodic_mlp_harmonic/backward/2026-07-08-01-19-22_te_periodic_mlp_harmonic_bw_simplified_setpoints/python/periodic_mlp-epoch=053-val_mae=0.00280310.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/periodic_mlp_harmonic/global/2026-07-08-01-02-50_te_periodic_mlp_harmonic_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/periodic_mlp_harmonic/global/2026-07-08-01-02-50_te_periodic_mlp_harmonic_global_simplified_setpoints/python/periodic_mlp-epoch=059-val_mae=0.00284742.ckpt</code>

Aggregate Metrics

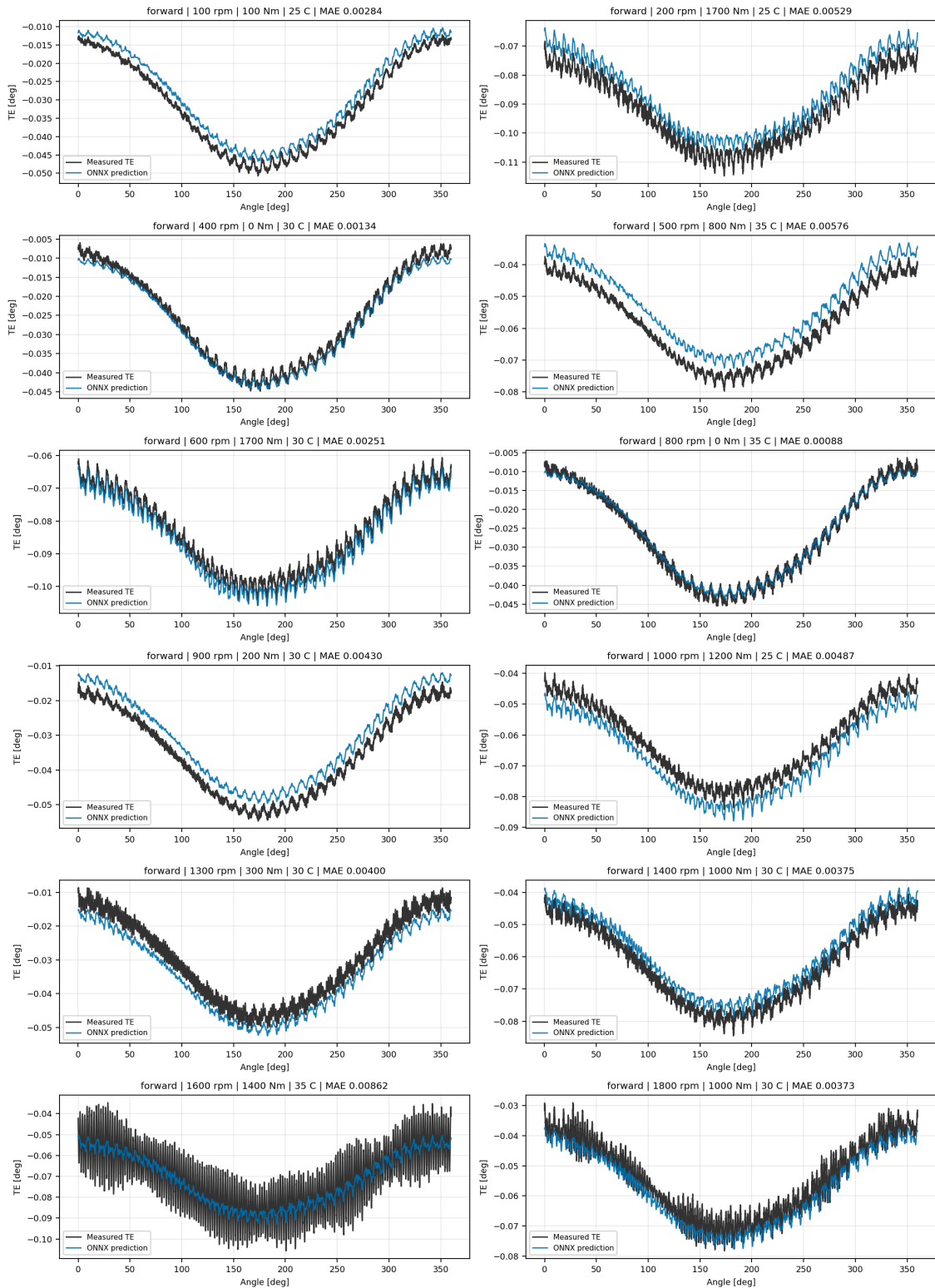
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003111	0.003363	6.899	11.570
backward	97	0.003842	0.004144	8.477	16.308
global	194	0.003515	0.003796	7.758	13.911

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000032	0.002726	0.001170	0.004345
backward	0.001274	0.003256	0.001501	0.005290
global	0.000233	0.003036	0.001344	0.005857

Forward 12-Curve Page

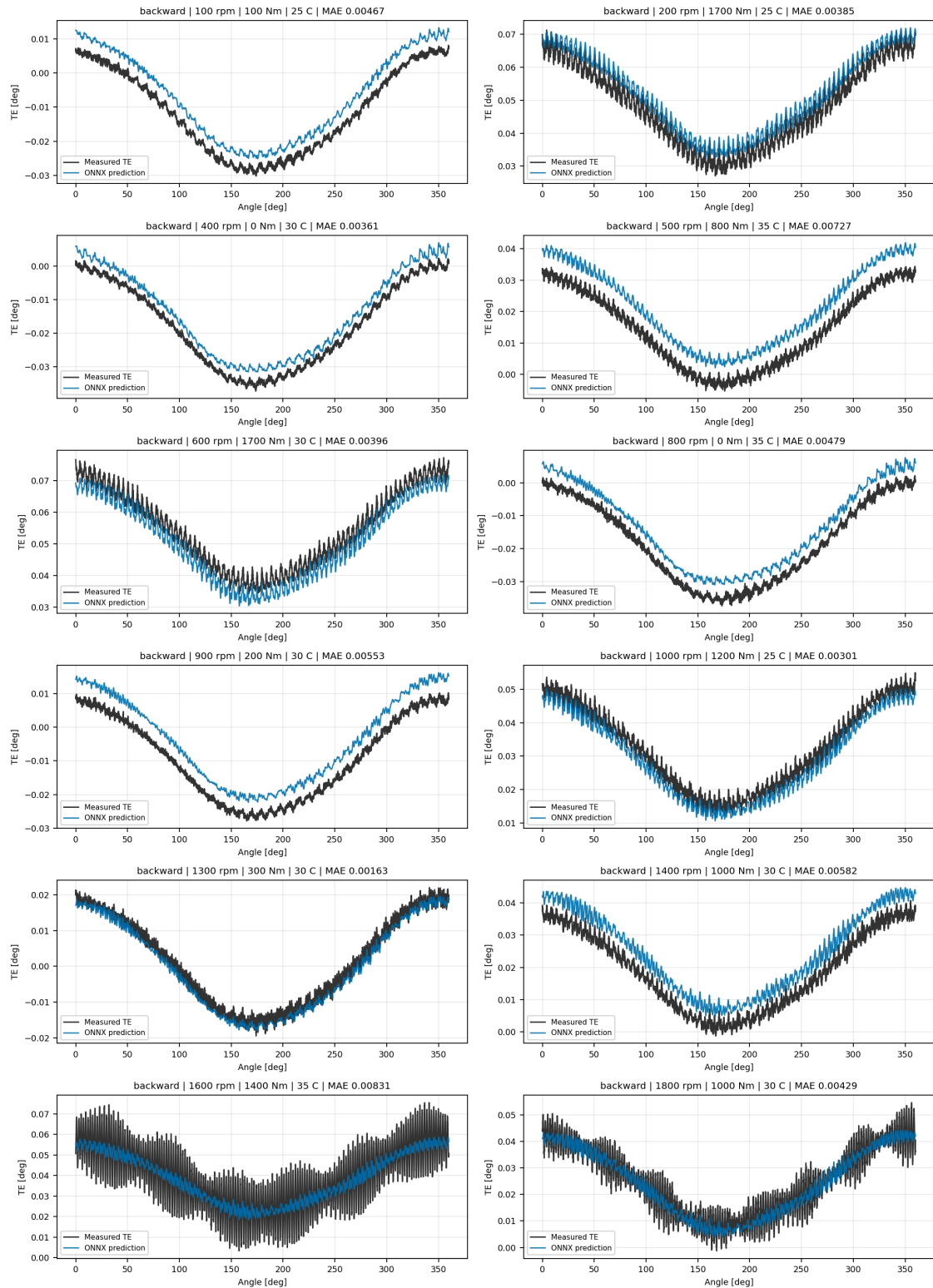
Simplified Dataset + Setpoints | periodic_mlp_harmonic | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

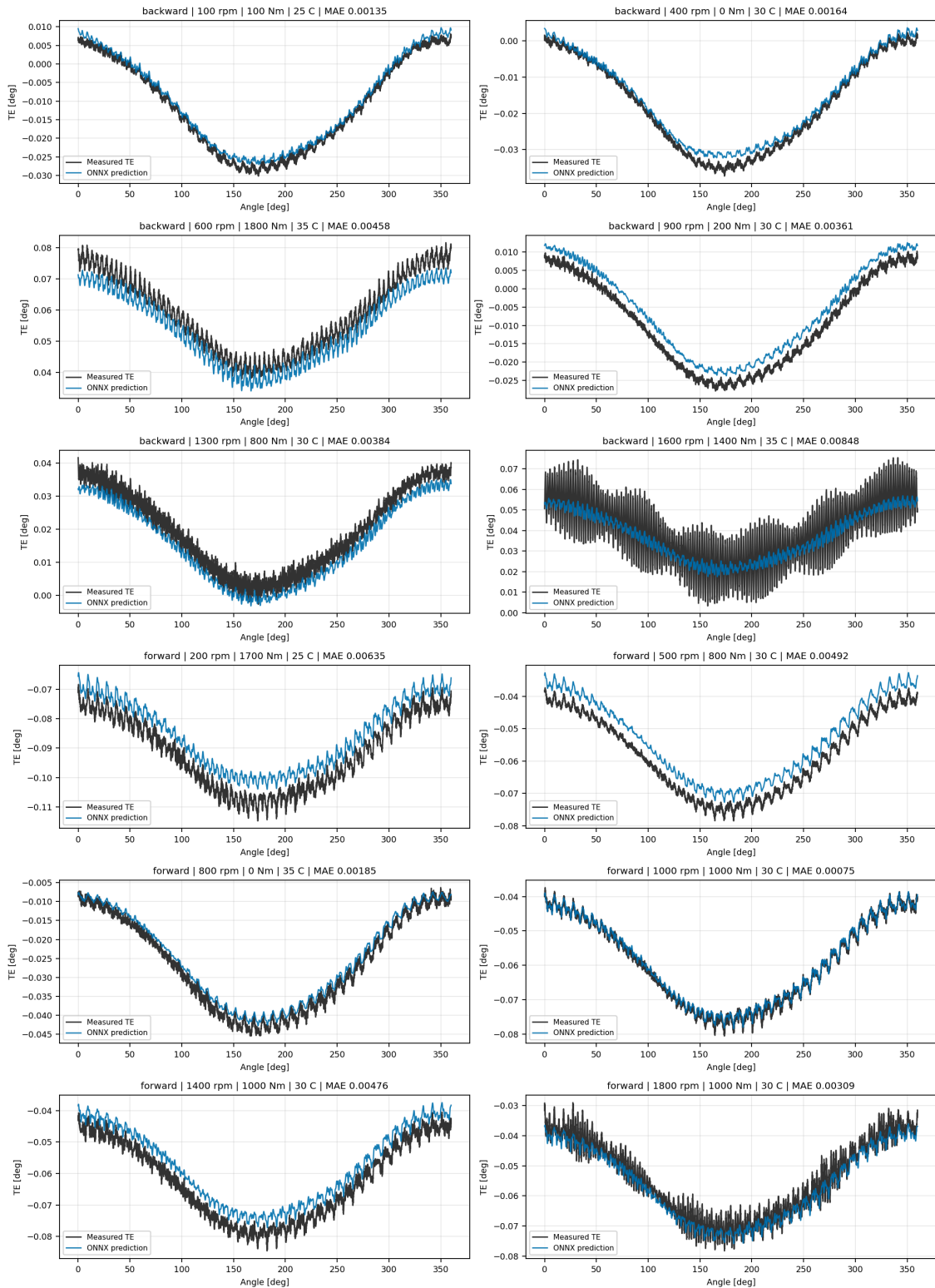
Simplified Dataset + Setpoints | periodic_mlp_harmonic | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | periodic_mlp_harmonic | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: periodic_mlp_harmonic ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_periodic_mlp_harmonic_fw_polished_setpoints	2026-07-08-02-15-59_te_periodic_mlp_harmonic_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_periodic_mlp_harmonic_bw_polished_setpoints	2026-07-08-02-29-32_te_periodic_mlp_harmonic_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_periodic_mlp_harmonic_global_polished_setpoints	2026-07-08-01-46-27_te_periodic_mlp_harmonic_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/periodic_mlp_harmonic/forward/2026-07-08-02-15-59_te_periodic_mlp_harmonic_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/periodic_mlp_harmonic/forward/2026-07-08-02-15-59_te_periodic_mlp_harmonic_fw_polished_setpoints/python/periodic_mlp-epoch=051-val_mae=0.00120808.ckpt
backward	models/polished_dataset/setpoints/exported/periodic_mlp_harmonic/backward/2026-07-08-02-29-32_te_periodic_mlp_harmonic_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/periodic_mlp_harmonic/backward/2026-07-08-02-29-32_te_periodic_mlp_harmonic_bw_polished_setpoints/python/periodic_mlp-epoch=081-val_mae=0.00121896.ckpt
global	models/polished_dataset/setpoints/exported/periodic_mlp_harmonic/global/2026-07-08-01-46-27_te_periodic_mlp_harmonic_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/periodic_mlp_harmonic/global/2026-07-08-01-46-27_te_periodic_mlp_harmonic_global_polished_setpoints/python/periodic_mlp-epoch=183-val_mae=0.00113740.ckpt

Aggregate Metrics

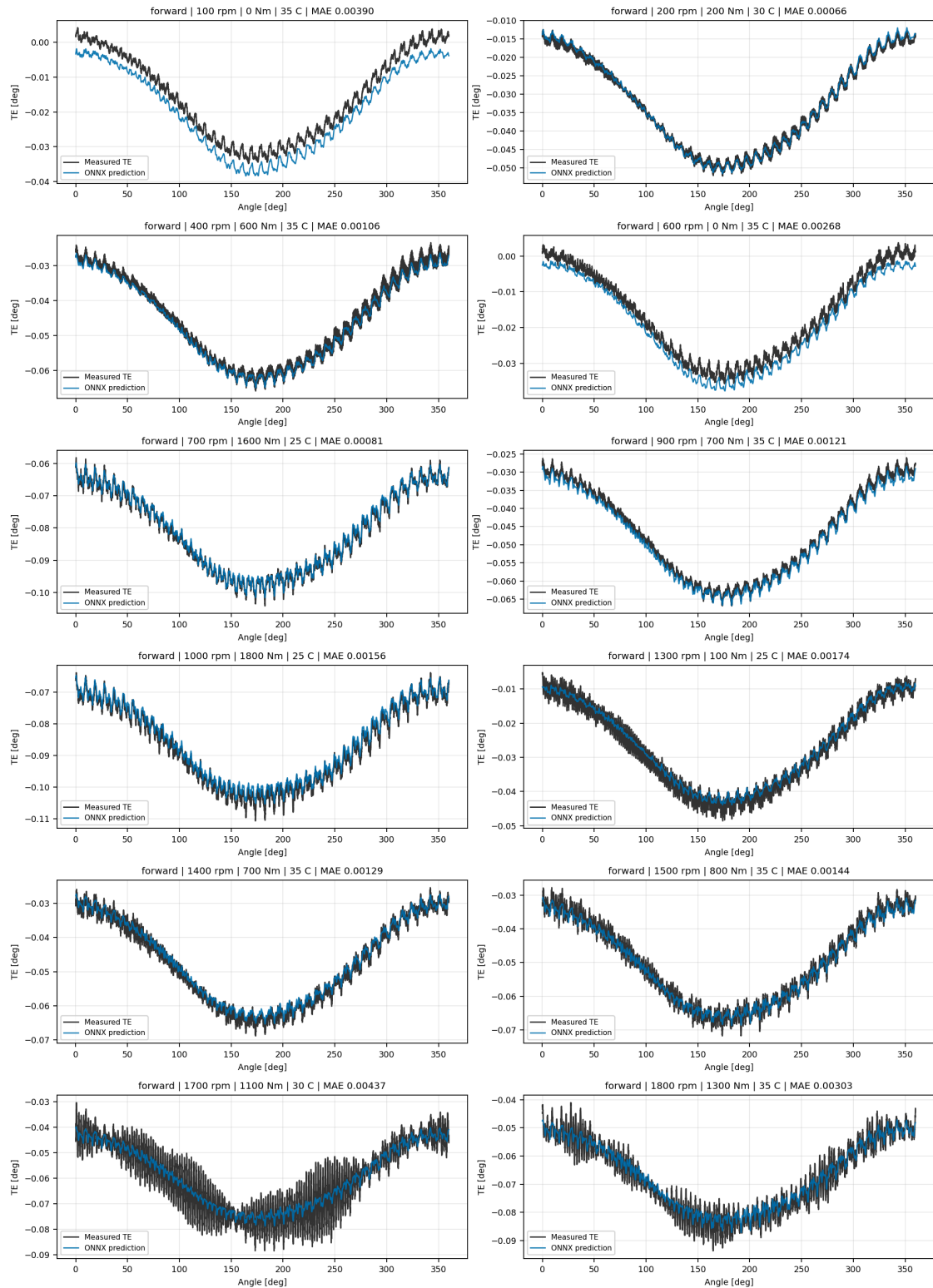
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001938	0.002276	3.988	10.017
backward	94	0.002470	0.002891	4.277	9.956
global	194	0.002113	0.002488	3.954	9.629

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000102	0.001008	0.001490	0.006883
backward	0.000137	0.000996	0.002026	0.008407
global	0.000078	0.000953	0.001726	0.006013

Forward 12-Curve Page

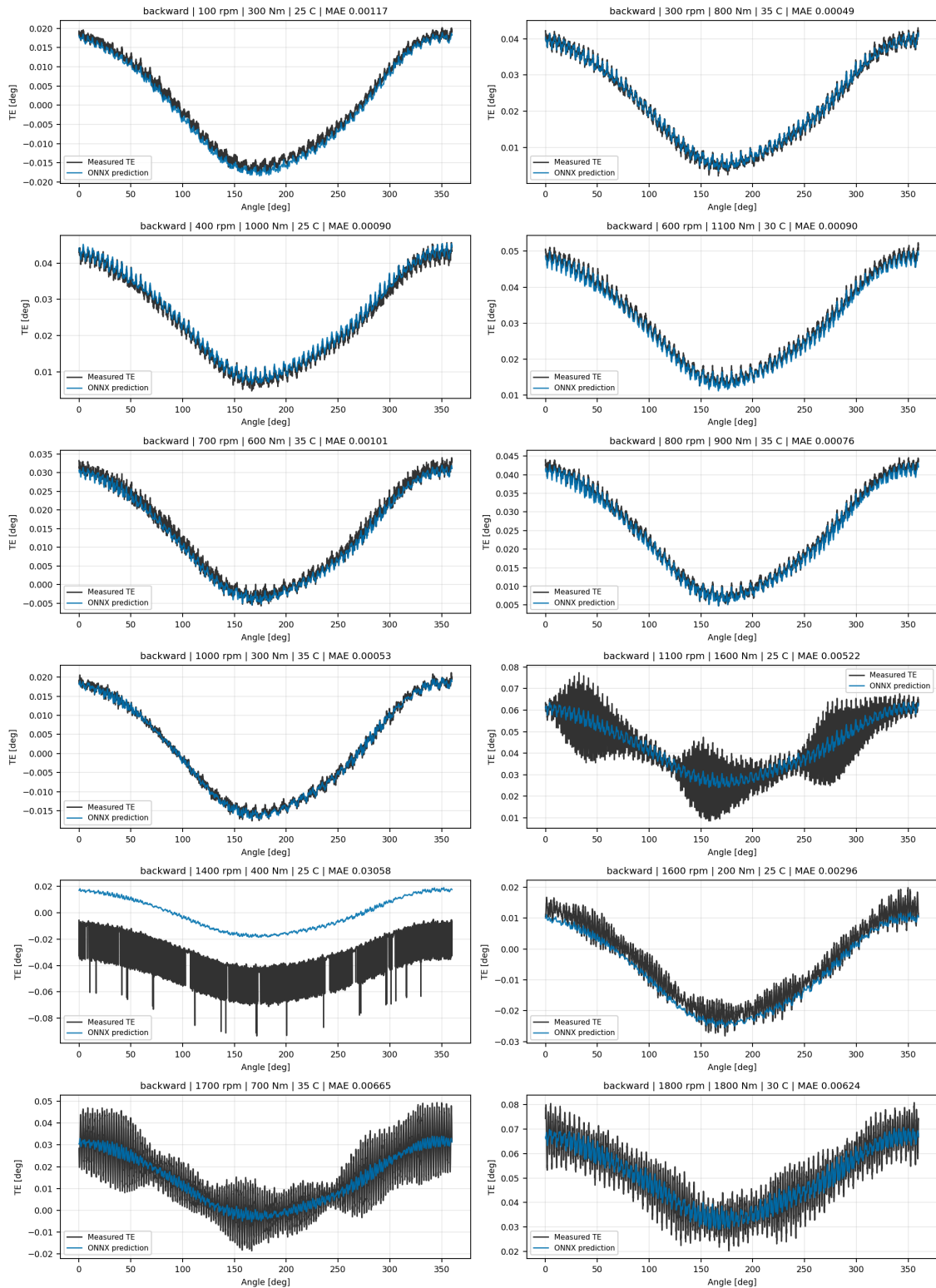
Polished Dataset + Setpoints | periodic_mlp_harmonic | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

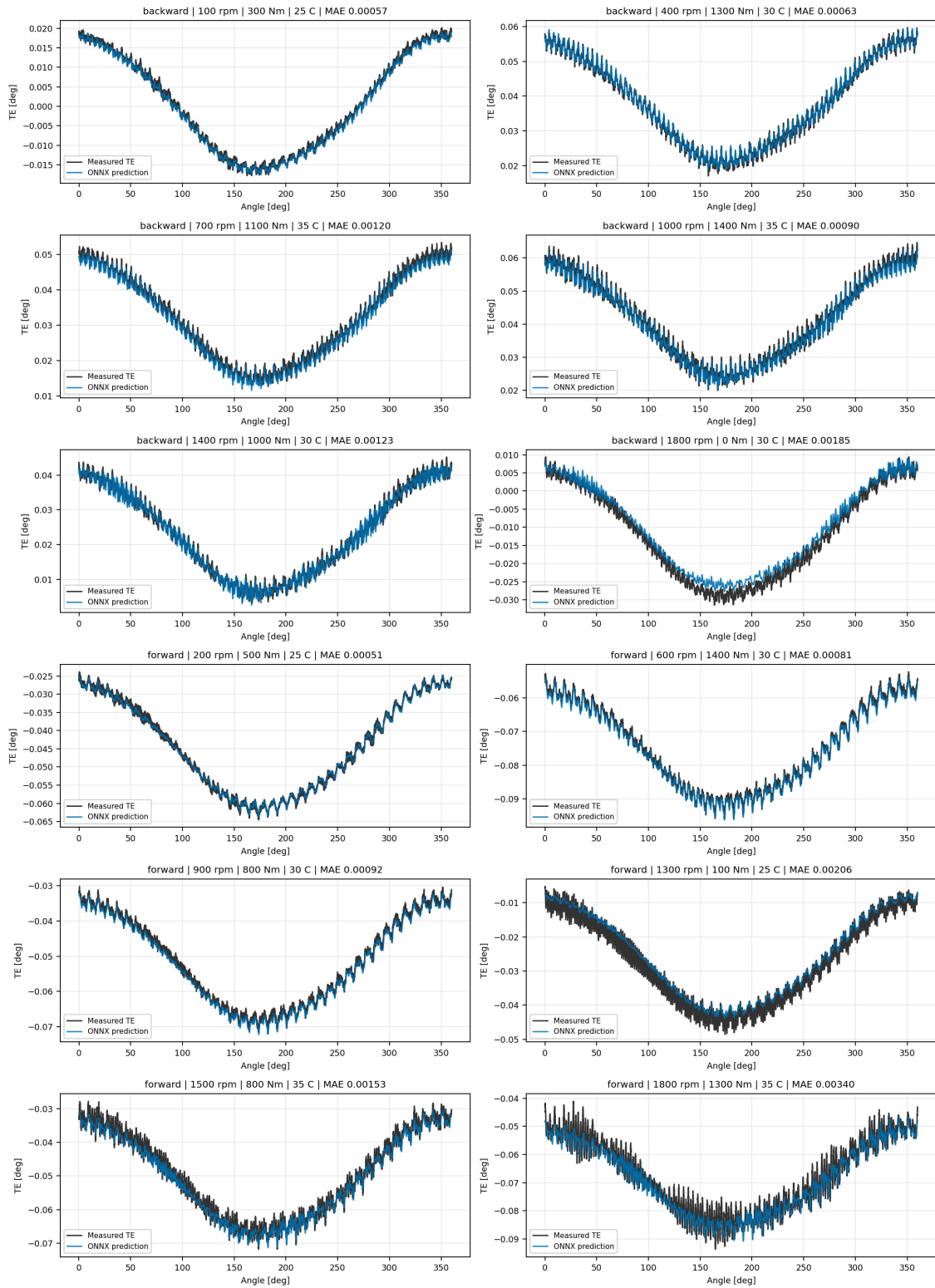
Polished Dataset + Setpoints | periodic_mlp_harmonic | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | periodic_mlp_harmonic | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: periodic_mlp_harmonic ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_periodic_mlp_harmonic_fw_polished_actual_values	2026-07-08-03-17-38_te_periodic_mlp_harmonic_fw_polished_actual_values	polished_point_v1
backward	te_periodic_mlp_harmonic_bw_polished_actual_values	2026-07-08-03-31-01_te_periodic_mlp_harmonic_bw_polished_actual_values	polished_point_v1
global	te_periodic_mlp_harmonic_global_polished_actual_values	2026-07-08-02-59-27_te_periodic_mlp_harmonic_global_polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/periodic_mlp_harmonic/forward/2026-07-08-03-17-38_te_periodic_mlp_harmonic_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/periodic_mlp_harmonic/forward/2026-07-08-03-17-38_te_periodic_mlp_harmonic_fw_polished_actual_values/python/periodic_mlp-epoch=045-val_mae=0.00131065.ckpt
backward	models/polished_dataset/actual_values/exported/periodic_mlp_harmonic/backward/2026-07-08-03-31-01_te_periodic_mlp_harmonic_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/periodic_mlp_harmonic/backward/2026-07-08-03-31-01_te_periodic_mlp_harmonic_bw_polished_actual_values/python/periodic_mlp-epoch=128-val_mae=0.00117146.ckpt
global	models/polished_dataset/actual_values/exported/periodic_mlp_harmonic/global/2026-07-08-02-59-27_te_periodic_mlp_harmonic_global_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/periodic_mlp_harmonic/global/2026-07-08-02-59-27_te_periodic_mlp_harmonic_global_polished_actual_values/python/periodic_mlp-epoch=074-val_mae=0.00123779.ckpt

Aggregate Metrics

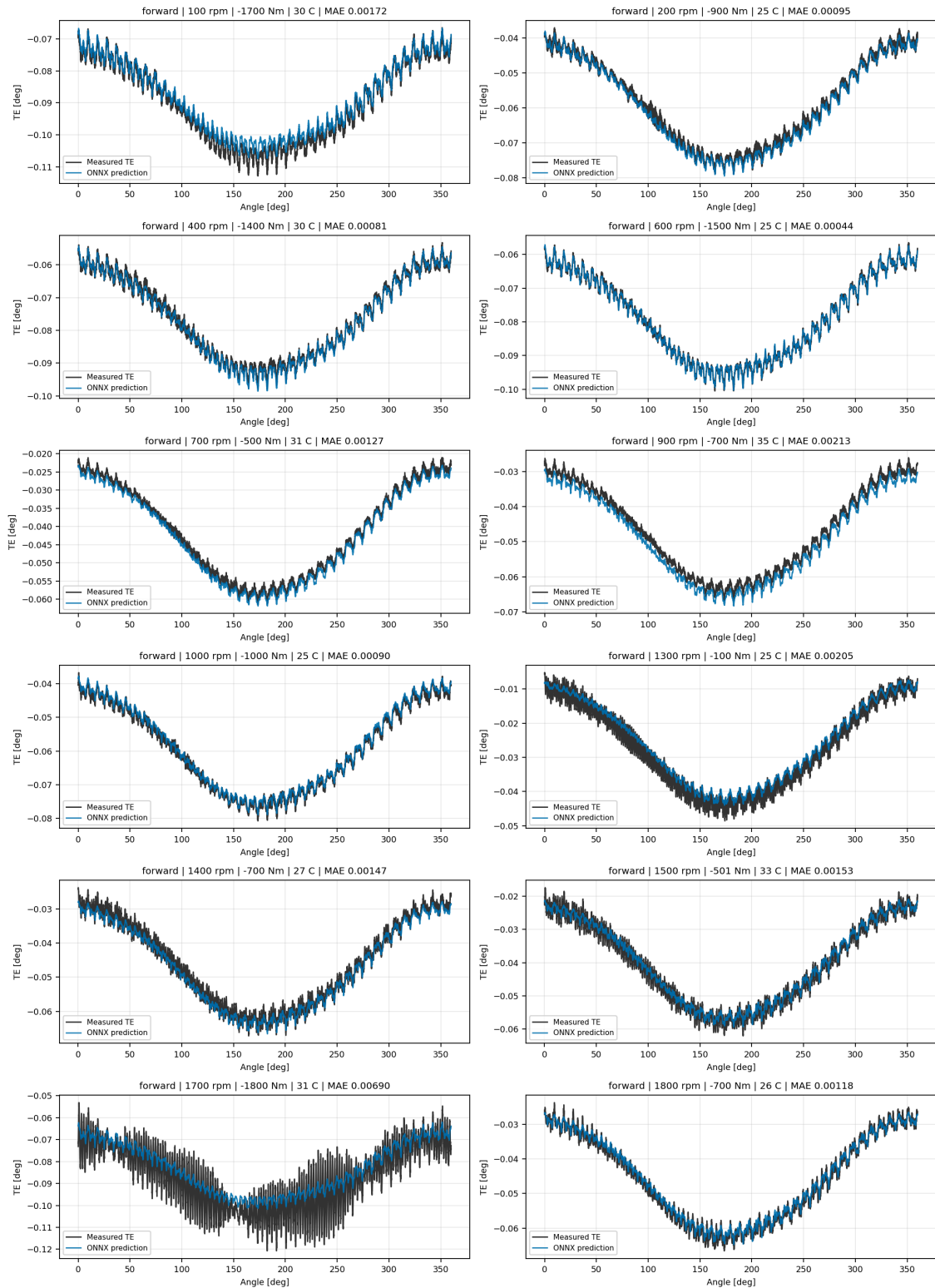
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001900	0.002239	3.891	10.013
backward	94	0.002483	0.002914	4.321	10.519
global	194	0.002153	0.002535	4.017	9.840

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000240	0.000943	0.001507	0.005793
backward	0.000503	0.000973	0.002072	0.007753
global	0.000146	0.000968	0.001756	0.006152

Forward 12-Curve Page

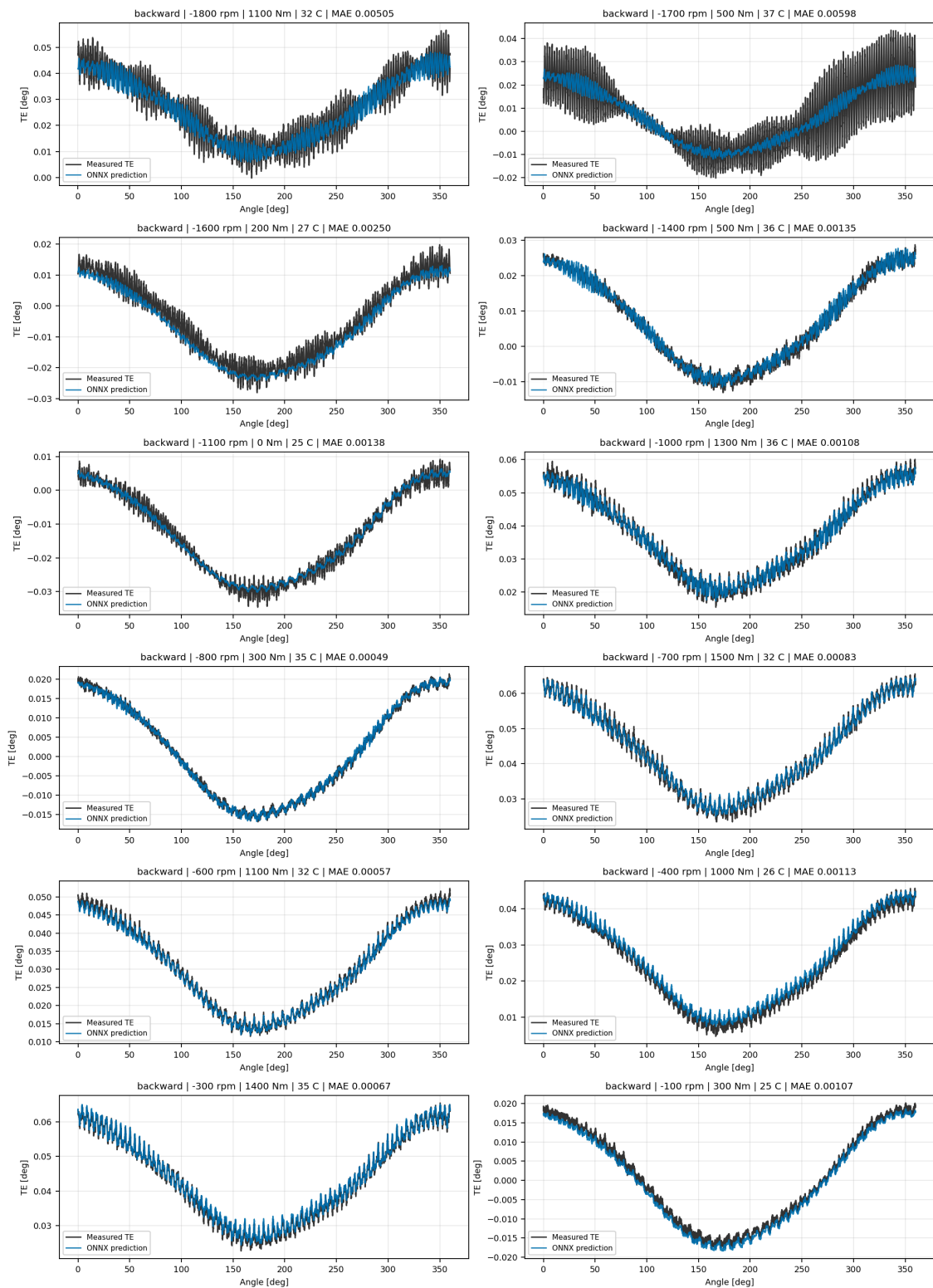
Polished Dataset + Actual Values | periodic_mlp_harmonic | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

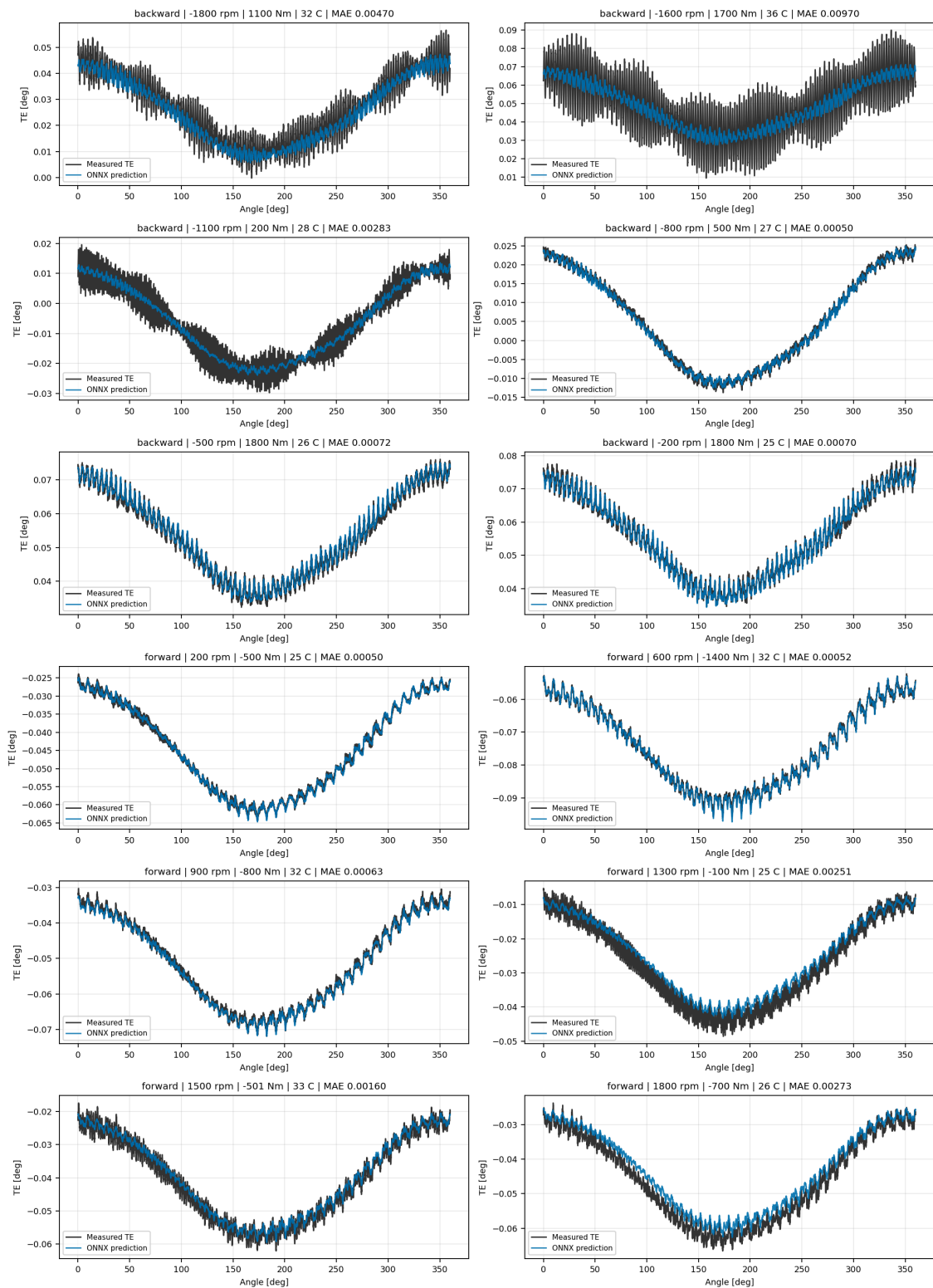
Polished Dataset + Actual Values | periodic_mlp_harmonic | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | periodic_mlp_harmonic | global



polished_dataset_actual_values global 12-curve collage